

EcoSphere Enterprise Sustainability & ESG Platform

Functional Requirements Specification
& Business Requirements Document

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Frameworks Covered: GRI 2021 | BRSR | TCFD | ISO 14001 | ISAE 3000 | MoEFCC EIA 2006

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Section 1 | Executive Summary

The **EcoSphere Enterprise Sustainability & ESG Platform** is a comprehensive digital solution designed to manage the complete sustainability lifecycle — from data collection at facility level to board-level disclosures. The platform is modelled on globally recognized sustainability reporting frameworks including **GRI Standards 2021**, **BRSR** (Business Responsibility and Sustainability Reporting), **TCFD** (Task Force on Climate-related Financial Disclosures), and **MoEFCC EIA Notification 2006**.

The platform enables organizations to:

- Collect sustainability data from all facilities, plants, and business units
- Manage evidence uploads with complete audit trails and version control
- Track progress against sustainability targets with RAG status indicators
- Generate publication-ready ESG, Sustainability, EIA, and Carbon reports
- Ensure regulatory compliance across environmental, social, and governance dimensions
- Enable board-level oversight and stakeholder transparency

Four Sustainability Pillars

Pillar	Name	Focus Areas
Pillar I	Ethical Governance Practices	Anti-corruption, board oversight, data privacy, regulatory compliance
Pillar II	Preserving Our Planet	Energy, GHG emissions, water, waste, biodiversity, air quality
Pillar III	Our People, Our Strength	Employee wellbeing, diversity, OHS, training, human rights
Pillar IV	Steering Towards Progress	Community development, supply chain, economic performance

Section 2 | Business Objectives

Ref	Objective	Description	Priority	Framework
BO-01	Centralized ESG Data Management	Single platform for all sustainability data across all facilities	Critical	GRI 2021
BO-02	Regulatory Compliance	Adherence to BRSR, GRI, TCFD, EIA, and local regulations	Critical	BRSR/GRI
BO-03	Target Tracking	Monitor progress against Net Zero 2045 and all sustainability targets	High	TCFD
BO-04	Stakeholder Reporting	Generate publication-ready reports for investors, regulators, customers	High	GRI 302-306
BO-05	Supply Chain Sustainability	Extend ESG requirements to suppliers and value chain partners	High	GRI 308/414
BO-06	Risk Management	Identify, assess, and mitigate climate and ESG risks	High	TCFD
BO-07	Evidence Management	Digital repository for all sustainability evidence with version control	Medium	ISO 14001
BO-08	Governance Oversight	Enable board and committee oversight of ESG performance	Critical	GRI 2-9
BO-09	Materiality Assessment	Conduct and document double materiality assessments	High	GRI 3
BO-10	Audit Readiness	Maintain audit-ready data with complete chain of custody	Critical	ISAE 3000

Section 3 | Platform Scope & Modules

Module #	Module Name	Pillar	Key Functions	Frameworks
M-01	Sustainability Dashboard	All	KPI cards, target progress, alerts, charts	All
M-02	Governance & Organization	Pillar I	Org profile, committee, charter, meetings	GRI 2-9
M-03	Materiality Assessment	All	18 topics, double materiality, stakeholder input	GRI 3
M-04	Stakeholder Engagement	All	Surveys, meetings, feedback, engagement log	GRI 2-29
M-05	Energy Management	Pillar II	Electricity, diesel, coal, CNG, renewables	GRI 302
M-06	Water Management	Pillar II	Freshwater, recycled, groundwater, effluent	GRI 303
M-07	Waste Management	Pillar II	Hazardous, non-hazardous, recycled, landfill	GRI 306
M-08	GHG & Emissions	Pillar II	Scope 1, 2, 3 emissions tracking	GRI 305/TCFD
M-09	Workforce & Diversity	Pillar III	Headcount, gender, hires, attrition, PwD	GRI 401/405
M-10	Occupational Health & Safety	Pillar III	LTI, near miss, fatalities, audits	GRI 403
M-11	Training & Development	Pillar III	Training hours, programs, coverage	GRI 404
M-12	Compliance & Governance	Pillar I	Policies, legal register, whistleblower	GRI 205/206
M-13	Supply Chain Sustainability	Pillar IV	Supplier registration, ESG scores, CoC	GRI 308/414
M-14	ESG Risk Management	All	Risk register, heat map, TCFD risk categories	TCFD
M-15	Reporting & Disclosures	All	ESG, EIA, Carbon, BRSR report generation	All Frameworks

Section 4 | Organization Structure

The platform supports a multi-level organizational hierarchy enabling data collection, review, and approval at each level:

Level	Entity Type	Examples	Sustainability Role
Level 1	Corporate Organization	Group / Holding Company	Strategic oversight, board governance, report publication
Level 2	Business Divisions	Pharmaceuticals, Chemicals, APIs	Division-level ESG targets, performance tracking
Level 3	Manufacturing Units/Plants	Plant A, Plant B, API Unit	Primary data collection, facility-level approvals
Level 4	R&D; Centers	Research Campus, Innovation Lab	R&D-specific; environmental & social data
Level 5	Departments	HR, HSE, Finance, Procurement	Module-specific data entry and evidence upload
Level 6	Employees & Contractors	Individual contributors	Training records, incident reporting, survey responses

Additional Entities Supported:

- Sustainability Committee — cross-functional body driving ESG agenda
- Board of Directors — ultimate ESG accountability and report sign-off
- External Suppliers — self-service supplier portal for ESG data submission
- External Stakeholders — investors, regulators, NGOs, local communities

Section 5 | User Roles & Approval Hierarchy

5.1 User Roles & Permissions

Role ID	Role Name	Level	Permissions	Approval Authority
R-01	Super Administrator	System	Full access, configuration, user management	All modules
R-02	Board Member	L1	Read-all, approve ESG reports, review governance	Final approval
R-03	Chief Sustainability Officer	L1	All modules, report publication, target setting	L1 approver
R-04	Sustainability Manager	L2	Data review, module management, report drafts	L2 approver
R-05	Sustainability Coordinator	L3	Data entry, evidence upload, module updates	L3 reviewer
R-06	Plant / Facility Head	L3	Facility data approval, local oversight	Facility approver
R-07	HSE Officer	L4	EHS data entry, incident logging, safety data	Data submitter
R-08	HR Manager	L4	Workforce data, training records, diversity data	Data submitter
R-09	Finance Officer	L4	Energy invoices, utility bills, capex data	Data submitter
R-10	Compliance Officer	L4	Policies, audits, legal register, whistleblower	Data submitter
R-11	Procurement Manager	L4	Supplier data, CoC tracking, supply chain	Data submitter
R-12	IT Administrator	System	System config, integrations, user access	Technical admin
R-13	Supplier Representative	External	Supplier portal, own ESG data upload	Self-service
R-14	External Auditor	External	Read-only verified data, evidence review	Verification only
R-15	Sustainability Ambassador	Cross-functional	Champion data, awareness, training facilitation	Advocate
R-16	Report Viewer	Read-only	View dashboards, download published reports	None

5.2 Six-Level Approval Hierarchy

Step	Actor	Action	Outcome
Step 1	HSE / HR / Finance Officer	Submit data with evidence attachments	Record in 'Pending Review' status
Step 2	Plant / Facility Head	Review and validate facility-level data accuracy	Approved → moves to Step 3; Rejected → back to submitter
Step 3	Sustainability Coordinator	Verify evidence quality and completeness	Verified → moves to Step 4; Clarification requested if needed
Step 4	Sustainability Manager	Cross-facility consolidation and data review	Approved → moves to Step 5; Escalated if discrepancies found
Step 5	Chief Sustainability Officer	Strategic alignment check and final approval	Final approval → Data locked for reporting
Step 6	Board of Directors	Annual ESG report acknowledgment and sign-off	Report published; board resolution recorded

Section 6 | Sustainability Governance Module

6.1 Functional Requirements

Req ID	Requirement	Description	Evidence Required	Approver
GOV-01	Organization Profile	Capture company name, CIN, industry, HQ, sites, reporting year, CEO details	Certificate of Incorporation	CSO
GOV-02	Sustainability Charter	Upload and manage sustainability charter with version control and sign-off	Charter PDF (signed)	Board
GOV-03	Committee Management	Register all committee members: name, designation, role, division, tenure	Appointment letters	CSO
GOV-04	Meeting Log	Record meetings: date, type, attendees, agenda, minutes, action items	Meeting minutes PDF	Sustainability Mgr
GOV-05	Governance Structure	Visual hierarchy: Board → Steering Committee → Ambassadors → Champions	Org chart PDF	CSO
GOV-06	ESG Oversight	Track board ESG review frequency, topics discussed, and decisions taken	Board resolution	Board
GOV-07	Sustainability Ambassadors	Register ambassadors per facility with contact details and focus area	Nomination form	Facility Head
GOV-08	Sustainability Champions	Register champions per department with training status and activities	Champion certificate	Sustainability Mgr
GOV-09	Annual Planning	Create and track annual sustainability plan with milestones and owners	Plan document PDF	CSO
GOV-10	Review Calendar	Schedule sustainability review and board meetings with reminders	Calendar invites	CSO

6.2 Governance Workflow

- Annual Planning: CSO creates sustainability plan → approved by Board
- Committee Setup: Members registered → roles assigned → charter uploaded
- Quarterly Reviews: Meetings scheduled → minutes recorded → actions tracked
- Ambassador Network: Ambassadors nominated per facility → training completed
- Champion Program: Champions registered per department → activities logged
- Board ESG Review: Quarterly ESG summary presented → board decisions recorded
- Annual Report Initiation: CSO triggers report generation at year end

Section 7 | Data Collection Framework

Every data submission across all modules must capture the following universal metadata fields. This ensures complete traceability, audit readiness, and evidence linkage for every reported metric.

7.1 Universal Evidence Capture Fields

Field	Type	Required	Description
Facility Name	Dropdown	Yes	Site / plant / unit name from master list
Department	Dropdown	Yes	Submitting department from org hierarchy
Reporting Period	Date range	Yes	Month / Quarter / Year selection
Data Owner	User lookup	Yes	Responsible person for this data entry
Evidence Files	File upload	Yes	Bills, reports, certificates — see 7.2
Upload Date	Auto-filled	Yes	System timestamp at upload
Approval Status	Status	Auto	Pending / Under Review / Approved / Rejected
Verification Status	Status	Auto	Unverified / Verified / Certified (ISAE 3000)
Reviewer Comments	Text area	No	Notes, queries, or instructions from reviewer
Version Number	Auto-integer	Auto	Increments on each re-submission
Audit Trail	System log	Auto	All actions timestamped with user, action, result

7.2 Supported Evidence File Types

File Category	Accepted Formats	Max Size	Use Case
Images	JPG, PNG, HEIC, WEBP	10 MB	Site photos, equipment, events, awareness programs
Documents	PDF, DOCX, DOC	25 MB	Reports, policies, certificates, meeting minutes
Spreadsheets	XLSX, CSV, XLS	20 MB	Meter readings, monitoring data, HR extracts
Utility Bills	PDF, JPG, PNG	15 MB	Electricity, water, gas, fuel invoices
Compliance Certs	PDF	10 MB	ISO 14001, ISO 45001, BIS, OHSAS certificates
Audit Reports	PDF, DOCX	25 MB	Internal / external audit findings and responses
Training Records	PDF, XLSX, JPG	20 MB	Attendance sheets, completion certificates, photos

Section 8 | Environmental Data Collection

8.1 Energy Management (GRI 302)

Parameter	Unit	Source	Evidence Required	Frequency	GRI Code
Grid Electricity	kWh	Utility bills	Electricity bills + meter readings	Monthly	GRI 302-1
Diesel Consumption	Litres	Fuel invoices	Delivery invoices + DG log	Monthly	GRI 302-1
Coal Consumption	MT	Purchase invoices	Weighbridge slips + invoices	Monthly	GRI 302-1
CNG Consumption	SCM	Gas bills	Gas meter readings + bills	Monthly	GRI 302-1
Renewable Energy Generated	kWh	Generation meter	Solar/wind generation logs	Monthly	GRI 302-1
Renewable Energy Purchased	kWh	REC certificates	REC purchase documents	Monthly	GRI 302-1
Energy Intensity	GJ/unit	Calculated	Production records	Monthly	GRI 302-3

8.2 Water Management (GRI 303)

Parameter	Unit	Source	Evidence Required	Frequency	GRI Code
Municipal Water	KL	Water bills	Utility bills	Monthly	GRI 303-3
Groundwater	KL	Borewell meter	Meter log / extraction permit	Monthly	GRI 303-3
Recycled Water	KL	ETP / STP	ETP operational records	Monthly	GRI 303-3
Rainwater Harvested	KL	Collection meter	Rainwater harvesting log	Monthly	GRI 303-3
Effluent Discharged	KL	Flow meter	ETP discharge logs / NOC	Monthly	GRI 303-4
pH of Effluent	pH	Lab test	NABL lab test report	Monthly	GRI 303-4
BOD	mg/L	Lab test	NABL lab test report	Monthly	GRI 303-4
COD	mg/L	Lab test	NABL lab test report	Monthly	GRI 303-4
Water Intensity	KL/unit	Calculated	Production records	Monthly	GRI 303-5

8.3 Waste Management (GRI 306)

Parameter	Unit	Source	Evidence Required	Frequency	GRI Code
Hazardous Waste Generated	MT	Manifest	TSDf manifest / Form 3 / Form 10	Monthly	GRI 306-3
Non-Hazardous Waste	MT	Weighbridge	Weighbridge slips	Monthly	GRI 306-4
Recycled Waste	MT	Vendor receipt	Recycler authorization certificate	Monthly	GRI 306-5
Waste to Landfill	MT	Disposal cert	Landfill operator receipt	Monthly	GRI 306-5
Biomedical Waste	KG	Incinerator log	CBWTF receipt	Monthly	GRI 306-3
E-Waste	KG	Manifest	PRO / E-waste authorization	Quarterly	GRI 306-3

8.4 GHG Emissions (GRI 305 / TCFD)

Parameter	Unit	Scope	Source	Evidence Required	Frequency	GRI Code
Diesel (Stationary)	tCO ₂ e	Scope 1	Fuel invoices	Delivery invoices	Monthly	GRI 305-1
CNG Combustion	tCO ₂ e	Scope 1	Gas bills	Gas meter + bills	Monthly	GRI 305-1
Coal Combustion	tCO ₂ e	Scope 1	Coal invoices	Coal invoices	Monthly	GRI 305-1
Grid Electricity	tCO ₂ e	Scope 2	Electricity bills	Electricity bills	Monthly	GRI 305-2
Business Travel	tCO ₂ e	Scope 3	Travel records	Flight/hotel bills	Quarterly	GRI 305-3
Supply Chain	tCO ₂ e	Scope 3	Supplier data	Supplier reports	Annual	GRI 305-3
GHG Intensity	tCO ₂ e/unit	—	Calculated	Production records	Monthly	GRI 305-4

Section 9 | Social Data Collection

9.1 Workforce & Diversity (GRI 401 / 405)

Parameter	Unit	Source	Evidence Required	Frequency
Total Employees	Count	HRMS	HRMS extract / payroll report	Monthly
Male Employees	Count	HRMS	HRMS extract	Monthly
Female Employees	Count	HRMS	HRMS extract	Monthly
Contract Workers	Count	Contractor records	Contractor register	Monthly
Workers with Disabilities	Count	HR records	PwD declaration forms	Monthly
New Hires	Count	HRMS	Offer letters / joining report	Monthly
Attrition	Count	HRMS	Exit records / separation list	Monthly
Women in Leadership	%	HRMS	Org chart extract	Quarterly
Avg Training Hours	Hrs/employee	LMS	LMS training records	Monthly

9.2 Occupational Health & Safety (GRI 403)

Parameter	Unit	Source	Evidence Required	Frequency
Lost Time Injuries (LTI)	Count	Incident register	Incident investigation reports	Monthly
Lost Working Days (LWD)	Days	Incident register	Medical certificates	Monthly
Near Miss Incidents	Count	Near miss register	Near miss reports	Monthly
First Aid Cases (FAC)	Count	First aid register	First aid treatment records	Monthly
Fatalities	Count	Incident register	Investigation report, FIR copy	Immediate
Man-hours Worked	Hours	Attendance system	Attendance records, timesheet	Monthly
Safety Audits Conducted	Count	Audit schedule	Audit reports with findings	Monthly
Safety Observations	Count	Observation cards	Observation logs	Monthly
LTIFR (calculated)	Rate	System calculated	—	Monthly
TRIFR (calculated)	Rate	System calculated	—	Monthly

9.3 Training & Development (GRI 404)

Training Category	Evidence Required	Tracking Fields
Safety Training	Attendance sheets, certificate scans	Hours, participants, coverage %, pass rate
Environmental Training	Attendance + assessment scores	Hours, participants, topics covered
Ethics & Anti-Corruption	Completion certificates, sign-off sheets	Coverage %, completion date, re-training due
Human Rights	Sign-off sheets, declarations	Coverage %, employees covered, contractors
Leadership Development	Program completion records	Participants, program name, duration
Technical Skills	Certification copies, assessment results	Certifications earned, skills updated
Sustainability Awareness	Attendance sheets + event photos	Reach, engagement, quiz scores

Section 10 | Governance & Compliance

10.1 Policy Library (11 Required Policies)

Policy #	Policy Name	Review Frequency	Evidence Required	GRI Reference
P-01	Sustainability Policy	Annual	Board-signed PDF	GRI 2-23
P-02	Environmental Policy	Annual	Signed PDF	GRI 302-306
P-03	Human Rights Policy	Annual	Signed PDF	GRI 406-412
P-04	Anti-Bribery & Corruption Policy	Annual	Signed PDF	GRI 205
P-05	Diversity & Inclusion Policy	Annual	Signed PDF	GRI 405
P-06	Occupational Health & Safety Policy	Annual	Signed PDF	GRI 403
P-07	Supplier Code of Conduct	Annual	Signed CoC document	GRI 308/414
P-08	Whistleblower Protection Policy	Annual	Signed PDF	GRI 2-26
P-09	Data Privacy Policy	Annual	Signed PDF	GRI 418
P-10	Community Engagement Policy	Biennial	Signed PDF	GRI 413
P-11	Climate Change Policy	Annual	Board-signed PDF	GRI 201/305

10.2 Legal Compliance Register

Field	Description	Options / Format
Act / Regulation	Name of applicable law or standard	Free text with search
Jurisdiction	Scope of the regulation	National / State / Local / International
Compliance Status	Current compliance position	Compliant / Non-Compliant / Partial / N/A
Due Date	Next compliance deadline or renewal	Date picker
Responsible Owner	User assigned for compliance action	User lookup dropdown
Evidence	Supporting document for compliance	NOC / Certificate / Consent order
Last Audit Date	Date of last compliance check	Date picker
Next Review Date	Scheduled next review	Auto-calculated (+ 6 or 12 months)

10.3 Whistleblower Case Management Workflow

Step 1: Case Registered — Ref number auto-generated; date, category, and description captured

Step 2: Investigator Assigned — Compliance Officer or designated investigator notified

Step 3: Investigation Initiated — Evidence collected; interviews conducted if required

Step 4: Status Updated — Progress tracked: Open → Under Investigation → Resolved

Step 5: Resolution Documented — Resolution summary, corrective actions, and evidence uploaded

Step 6: Case Closed — Closure date recorded; complainant notified (if not anonymous)

Step 7: Annual Summary — All cases summarized for board ESG review

Section 11 | Stakeholder Engagement

11.1 Stakeholder Groups & Engagement Methods

Group #	Stakeholder Group	Primary Concerns	Engagement Methods
SG-01	Investors & Shareholders	ESG ratings, financial risks, disclosures	Annual reports, AGM, ESG briefings
SG-02	Employees & Unions	Safety, wellbeing, fair wages, growth	Surveys, town halls, grievance mechanism
SG-03	Customers	Product safety, sustainability credentials	Audits, sustainability certificates, meetings
SG-04	Suppliers & Vendors	CoC compliance, payment terms, ESG audit	Supplier portal, workshops, audits
SG-05	Regulators & Government	Compliance, reporting accuracy, permits	Regulatory filings, inspections, meetings
SG-06	Local Communities	Pollution, employment, CSR programs	Community meetings, CSR events, surveys
SG-07	NGOs & Civil Society	Environmental impact, human rights	Partnerships, consultations, reports
SG-08	Media	Transparency, positive ESG story	Press releases, ESG report launch events
SG-09	Industry Associations	Sector standards, benchmarking	Industry forums, working groups
SG-10	Academic Institutions	Research collaboration, internships	MoUs, internship programs, joint research

11.2 Materiality Assessment Process (Double Materiality)

- Step 1: Identify Material Topics — Pre-loaded 18 topics from GRI Universal Standards
- Step 2: Stakeholder Surveys — Collect impact materiality scores (1–5 scale) from all stakeholder groups
- Step 3: Financial Materiality Scoring — Internal team scores financial impact potential (1–5 scale)
- Step 4: Double Materiality Matrix — Plot all topics on impact vs. financial materiality axes
- Step 5: Prioritize Top Topics — Identify high-priority topics based on combined score threshold
- Step 6: Map to GRI Disclosures — Each material topic linked to applicable GRI topic standards
- Step 7: Board Approval — Final materiality matrix reviewed and approved by Board / Sustainability Committee

11.3 18 Material Topics

Topic #	Material Topic	Pillar	GRI Code	Impact Type
T-01	Ethical Business Practices	Pillar I	GRI 205	Impact + Financial
T-02	Regulatory Compliance	Pillar I	GRI 207	Financial
T-03	ESG Governance	Pillar I	GRI 2-9	Impact + Financial
T-04	Data Privacy & Security	Pillar I	GRI 418	Financial
T-05	Energy Management	Pillar II	GRI 302	Impact + Financial
T-06	GHG Emissions & Climate	Pillar II	GRI 305	Impact + Financial
T-07	Water Stewardship	Pillar II	GRI 303	Impact + Financial
T-08	Waste & Circular Economy	Pillar II	GRI 306	Impact
T-09	Biodiversity	Pillar II	GRI 304	Impact
T-10	Air Quality	Pillar II	GRI 305	Impact + Financial
T-11	Employee Wellbeing	Pillar III	GRI 401	Impact + Financial
T-12	Diversity & Inclusion	Pillar III	GRI 405	Impact + Financial

Topic #	Material Topic	Pillar	GRI Code	Impact Type
T-13	Occupational Health & Safety	Pillar III	GRI 403	Impact + Financial
T-14	Training & Development	Pillar III	GRI 404	Impact + Financial
T-15	Human Rights	Pillar III	GRI 406-412	Impact
T-16	Community Development	Pillar IV	GRI 413	Impact
T-17	Supply Chain Sustainability	Pillar IV	GRI 308/414	Impact + Financial
T-18	Economic Performance	Pillar IV	GRI 201	Financial

Section 12 | Supplier Sustainability

12.1 Supplier Registration & ESG Data Requirements

Field	Description	Evidence Required	Required
Code of Conduct Signed	Digital acceptance of Supplier CoC	Signed CoC document	Mandatory
ISO 14001 Certification	Environmental management certification	Valid certificate copy	Mandatory
ISO 45001 Certification	OHS management certification	Valid certificate copy	Preferred
Carbon Footprint	Annual Scope 1 + Scope 2 emissions	Emissions report / data sheet	Mandatory
Water Consumption	Annual total water consumption	Utility bills summary	Mandatory
Waste Generated	Annual hazardous + non-hazardous waste	Disposal manifests	Mandatory
Employee Count	Total workforce including contractors	HR summary report	Mandatory
Safety Record	LTI count, fatalities, LTIFR	Incident log	Mandatory
ESG Audit Date	Date of last sustainability audit	Audit report	Preferred
Risk Level	High / Medium / Low classification	Risk self-assessment form	Mandatory

12.2 Supplier ESG Scoring Matrix (0–100 Scale)

Scoring Category	Weight	Max Score	Criteria
Environmental Performance	30%	30	Emissions, water use, waste recycling rate, ISO 14001
Social Performance	25%	25	Safety record, employee diversity, training, human rights
Governance & Compliance	25%	25	Anti-corruption policy, compliance status, audit findings
CoC Acceptance	10%	10	Signed and dated Supplier Code of Conduct
Certification Status	10%	10	ISO 14001, ISO 45001, BIS, and other applicable certs
TOTAL	100%	100	Score bands: 80-100 = Green, 50-79 = Amber, <50 = Red (High Risk)

Section 13 | Targets & Goals Tracking

Target ID	Target Description	Baseline	Target Value	Target Year	Status	Owner
TGT-01	Achieve Net Zero GHG Emissions	2019 baseline	Net Zero	2045	On Track	CSO
TGT-02	Reduce Scope 1 & 2 by 25%	2019 baseline	-25%	2030	On Track	Energy Mgr
TGT-03	25% Renewable Energy Share	5% (2019)	25%	2030	On Track	Facilities
TGT-04	Reduce Freshwater Consumption 33%	2019 baseline	-33%	2030	At Risk	Water Mgr
TGT-05	Zero Waste to Landfill	Current MT	0 MT	2026	On Track	HSE Officer
TGT-06	30% Women in Workforce	Current %	30%	2030	On Track	HR Manager
TGT-07	100% Supplier CoC Acceptance	Current %	100%	2026	On Track	Procurement
TGT-08	100% Human Rights Training Coverage	Current %	100%	2025	Completed	HR Manager
TGT-09	Zero Fatalities	0 target	0	Annual	On Track	HSE Officer
TGT-10	20% Increase in Training Hours	Baseline hrs	+20%	2025	On Track	L&D; Head
TGT-11	100% Facilities ISO 14001 Certified	Count	100%	2026	On Track	EHS Head
TGT-12	50% Recycled Water Usage	Current %	50%	2030	At Risk	Water Mgr

Section 14 | ESG Risk Management

14.1 Risk Register Fields

Field	Type	Description
Risk ID	Auto-generated	Unique identifier: RSK-001, RSK-002 etc.
Risk Description	Free text	Clear description of the risk event
Category	Dropdown	Physical / Transition / Compliance / Supply Chain / Safety
Likelihood (1-5)	Integer	1 = Rare, 2 = Unlikely, 3 = Possible, 4 = Likely, 5 = Almost Certain
Impact (1-5)	Integer	1 = Negligible, 2 = Minor, 3 = Moderate, 4 = Major, 5 = Catastrophic
Risk Score	Calculated	Likelihood × Impact = Score (1–25); Heat map color coded
Mitigation Plan	Text	Actions to reduce likelihood or impact
Owner	User lookup	Assigned risk owner responsible for mitigation
Review Date	Date	Next scheduled risk review date
Status	Dropdown	Open / Under Mitigation / Residual Risk / Closed
Evidence	File upload	Supporting documents for risk assessment

14.2 TCFD Risk Categories & Pre-loaded Risks

Risk ID	Risk Description	Category	Likelihood	Impact	Score	Mitigation Owner
HR-01	Carbon pricing / carbon tax introduction	Transition	4	5	20	CSO
HR-02	Water scarcity affecting operations	Physical	4	4	16	Water Mgr
HR-03	Regulatory non-compliance (EIA/BRSR/GRI)	Compliance	3	5	15	Compliance Officer
HR-04	Supply chain ESG failures	Supply Chain	3	4	12	Procurement
HR-05	Extreme weather disruption to facilities	Physical	3	4	12	Facility Head
HR-06	Shift to stricter emission regulations	Transition	4	3	12	Energy Mgr

Section 15 | Reporting Framework

15.1 Report Types Supported

Report #	Report Name	Framework	Frequency	Approver
R-01	Annual Sustainability Report	GRI 2021	Annual	Board
R-02	BRSR Report	SEBI BRSR	Annual	CFO / CSO
R-03	Environmental Impact Assessment	MoEFCC EIA 2006	Project-based	Board / Regulator
R-04	Carbon Emissions Report	GRI 305 / TCFD	Annual	CSO
R-05	Water Stewardship Report	GRI 303	Annual	CSO
R-06	Waste Management Report	GRI 306	Annual	CSO
R-07	Safety Performance Report	GRI 403	Annual	HSE Head
R-08	Supplier ESG Report	GRI 308/414	Annual	Procurement
R-09	Board ESG Summary	Internal	Quarterly	Board
R-10	Facility-wise ESG Report	Internal	Monthly	Facility Head
R-11	Third-party Assurance Report	ISAE 3000	Annual	External Auditor

15.2 Annual Sustainability Report — Content Structure

Section #	Section Title	Data Source Module	Frameworks Mapped
1	Cover Page (company, year, pillar icons)	Governance Module	GRI 2-1
2	About the Report (boundary, scope, restatements)	Governance Module	GRI 2-2, 2-3
3	Message from Leadership (CEO, Chairman)	Governance Module	GRI 2-22
4	About the Organization	Governance Module	GRI 2-1 to 2-6
5	Sustainability Governance Structure	Governance Module	GRI 2-9 to 2-21
6	Materiality Assessment & Topics	Materiality Module	GRI 3-1, 3-2, 3-3
7	Stakeholder Engagement Summary	Stakeholder Module	GRI 2-29, 2-30
8	Environmental Performance	Energy/Water/Waste/GHG	GRI 302, 303, 305, 306
9	Social Performance	Workforce/OHS/Training	GRI 401, 403, 404, 405
10	Governance Performance	Compliance Module	GRI 205, 206, 418
11	Supply Chain Sustainability	Supply Chain Module	GRI 308, 414
12	Targets & Commitments (RAG status)	Targets Module	TCFD, BRSR
13	GRI Content Index	System auto-generated	GRI 2-55
14	Independent Assurance Statement	Evidence Repository	ISAE 3000

Section 16 | Evidence Management Framework

16.1 Evidence Lifecycle Stages

Stage	Stage Name	Action	Actor	Outcome
1	Upload	User uploads file with metadata (facility, period, type, description)	Data Submitter	File in Pending Review
2	Tagging	System auto-tags by module, period, facility; user adds manual tags	System + User	Fully tagged evidence record
3	Review	Reviewer opens evidence, comments, requests clarification if needed	Facility Head	Queries raised or passed to Step 4
4	Verification	Coordinator verifies evidence against reported data values	Sust. Coordinator	Verified status applied
5	Approval	Sustainability Manager approves evidence for use in reports	Sust. Manager	Approved — locked for reports
6	Archive	Approved evidence archived with 7-year retention policy	System	Immutable archive record
7	Audit Access	External auditors get read-only access to approved evidence	External Auditor	Audit trail accessible

16.2 Evidence Metadata Schema

Field	Type	Auto / Manual	Description
Evidence ID	String	Auto	System-generated unique identifier
File Name	String	Auto	Original file name at upload
File Type	String	Auto	PDF, XLSX, JPG, PNG, DOCX etc.
File Size	Integer	Auto	File size in KB
Upload Date	DateTime	Auto	Timestamp of upload
Uploaded By	User ref	Auto	User ID and name
Facility	Dropdown	Manual	Site/plant where evidence originated
Department	Dropdown	Manual	Submitting department
Module	Dropdown	Manual	Energy, Water, Safety, Training etc.
Reporting Period	Date range	Manual	Month/Quarter/Year coverage
Description	Text	Manual	Brief description of evidence content
Tags	Multi-tag	Manual	Searchable keyword tags
Approval Status	Enum	Auto	Pending / Under Review / Approved / Rejected
Approved By	User ref	Auto	Approver's user ID and name
Approval Date	DateTime	Auto	Timestamp of approval action
Linked Data Records	List	Auto	Linked data entries in any module

Section 17 | Screen Requirements

SCR-01: Sustainability Dashboard

- Organization KPI summary: 6 cards — Total GHG Emissions, Total Energy, Total Water, Total Waste, LTI Count, Women %
- Target progress bars: 12 targets with RAG color coding (Green $\geq$ target, Amber 70–99%, Red $<$ 70%)
- Module completion status: data collection progress per module per period
- Alerts panel: overdue approvals, missing data submissions, upcoming deadlines
- Quick action buttons: Add Data, Upload Evidence, Generate Report
- Interactive charts: GHG trend (line chart), Energy mix (pie), Water balance (bar), Waste composition (donut)

SCR-02: Data Entry Forms (All Modules)

- Header: module name, facility selector (dropdown), reporting period selector, data owner field
- Data grid: Parameter | Unit | Value field | Trend indicator | Previous period | % change
- Evidence upload zone: drag-and-drop interface, file type validation, max 25 MB per file
- Notes / comments free text field
- Save as Draft / Submit for Review action buttons
- Mandatory field validation with clear error messages before submission

SCR-03: Evidence Repository

- Search and filter bar: by module, facility, period, approval status, file type
- Grid / list toggle view
- Evidence cards showing: thumbnail, file name, facility, period, status badge, upload date
- Bulk actions: Download selected, Approve selected, Reject selected
- Evidence detail panel: file preview, metadata, approval history, comments thread

SCR-04: Approval Workflow Screen

- My Approvals queue: pending items sorted by due date with module and facility context
- Data summary: submitted values vs. previous period comparison with trend indicator
- Evidence preview panel: side-by-side with data values
- Action buttons: Approve / Reject / Request Clarification
- Comments field: mandatory on rejection
- Escalation button: send to next level with escalation reason
- Approval history timeline: all actions with timestamp and user name

SCR-05: Report Generation Screen

- Report type selector: dropdown of all 11 report types
- Reporting period selector: Year / Quarter / Month
- Facility scope selector: All facilities / Specific sites (multi-select)
- Data completeness indicator: % of data approved for the selected period
- Section customization: toggle individual report sections on/off
- Preview panel: live PDF preview before download

- Download button: generates final PDF
- Share options: email recipient list, shareable link
- Report history: all previously generated reports with version numbers

Section 18 | User Workflows

18.1 Workflow 1: Monthly Environmental Data Submission

Actor: HSE Officer (Plant Level) | Trigger: Start of new month | Frequency: Monthly

Step	Action
Step 1	Login to EcoSphere → Navigate to Energy module → Select facility and reporting month
Step 2	Enter grid electricity (kWh), diesel (litres), coal (MT), CNG (SCM), renewable energy (kWh)
Step 3	Upload utility bills: electricity bill PDF + meter reading image
Step 4	Navigate to Water module → Enter municipal, groundwater, recycled, effluent volumes
Step 5	Upload water bills and ETP monitoring reports
Step 6	Navigate to Waste module → Enter hazardous, non-hazardous, recycled, landfill quantities
Step 7	Upload TSDF manifests and weighbridge slips for waste data
Step 8	System auto-calculates GHG emissions using approved emission factors
Step 9	Review calculated values against previous month (trend alert if >20% deviation)
Step 10	Click 'Submit for Review' → System sends notification to Plant/Facility Head
Step 11	Facility Head logs in → Reviews data → Approves or rejects with mandatory comments
Step 12	Sustainability Coordinator receives notification → Verifies evidence documents
Step 13	Sustainability Manager reviews cross-facility data → Provides final approval
Step 14	Data locked for reporting period → Available for automatic report population

18.2 Workflow 2: Annual ESG Report Generation

Actor: Sustainability Manager + CSO | Trigger: End of reporting financial year | Frequency: Annual

Step	Action
Step 1	CSO initiates Annual Report in Reporting module → Selects FY and organizational scope
Step 2	System checks data completeness → Flags modules / facilities with missing approvals
Step 3	Sustainability Coordinator resolves all data gaps and completes pending approvals
Step 4	System auto-populates all data sections from approved records in all modules
Step 5	Materiality matrix auto-pulled from Materiality module with scores and ranking
Step 6	Target progress auto-pulled from Targets module (RAG status auto-calculated)
Step 7	Stakeholder engagement summary auto-pulled from Stakeholder module
Step 8	GRI Content Index auto-generated with all applicable disclosures mapped
Step 9	Draft report generated as PDF → Sustainability Manager reviews all sections
Step 10	CEO message entered or approved → Chairman/MD message entered
Step 11	Third-party assurance firm given read-only access to verified data set
Step 12	Assurance statement PDF uploaded by external auditor
Step 13	Final PDF generated with all sections, charts, images, and assurance statement
Step 14	Board review session → Board acknowledges via digital signature in platform
Step 15	Report published → Downloadable from platform, shared with stakeholders and website

18.3 Workflow 3: Supplier ESG Onboarding

Actor: Procurement Manager + Supplier Representative | Trigger: New supplier registration | Frequency: As required

Step	Action
Step 1	Procurement Manager creates supplier profile in Supply Chain module with basic details
Step 2	System sends email invitation to supplier with secure portal registration link
Step 3	Supplier registers on portal: company details, contact information, GST/PAN
Step 4	Supplier reviews and digitally accepts the Supplier Code of Conduct (mandatory)
Step 5	Supplier uploads ISO certificates, latest sustainability report, and emissions data
Step 6	Supplier completes ESG questionnaire (environmental, social, governance sections)
Step 7	System auto-calculates Supplier ESG Score (0–100) based on scoring matrix
Step 8	Procurement Manager reviews supplier submission and scoring results
Step 9	Risk classification assigned: High / Medium / Low based on score and category
Step 10	High-risk suppliers flagged → On-site ESG audit scheduled within 30 days
Step 11	Supplier profile activated → Available for procurement team decisions
Step 12	Annual re-assessment reminder sent automatically based on onboarding anniversary

Section 19 | Non-Functional Requirements

NFR ID	Category	Requirement	Specification
NFR-01	Performance	Dashboard page load time	< 3 seconds on standard broadband connection
NFR-02	Performance	Annual PDF report generation	< 30 seconds for full sustainability report
NFR-03	Security	Authentication	Multi-factor authentication (MFA) mandatory for all users
NFR-04	Security	Data encryption	AES-256 at rest; TLS 1.3 in transit
NFR-05	Security	Role-based access control	RBAC with least-privilege principle; no default open access
NFR-06	Availability	Platform uptime SLA	99.9% uptime SLA (< 8.76 hours unplanned downtime/year)
NFR-07	Scalability	Concurrent users	Support 500+ simultaneous users without degradation
NFR-08	Storage	Evidence file storage	Minimum 5 TB scalable cloud storage with expansion capability
NFR-09	Audit Trail	Activity logging	Immutable audit log: all actions timestamped with user and result
NFR-10	Retention	Data retention policy	7 years for all sustainability data and evidence files
NFR-11	Compliance	Data residency	India data residency for DPDP Act 2023 compliance
NFR-12	Accessibility	WCAG 2.1 standard	AA level accessibility compliance for all screens
NFR-13	Mobile	Responsive design	Mobile-friendly for field data entry (tablet / phone)
NFR-14	Integration	ERP system connectors	SAP, Oracle, Tally integration-ready APIs for master data sync
NFR-15	Backup	Disaster recovery	Daily backups; 4-hour RTO; 1-hour RPO for critical data

Section 20 | Document Control

Version	Date	Author / Team	Changes Made
v0.1	January 2026	EcoSphere Platform Team	Initial draft — core modules defined
v0.2	March 2026	Sustainability Team	Added supplier sustainability module, scoring matrix
v0.3	May 2026	Governance Review Team	Added governance workflows, approval hierarchy
v1.0	June 2026	CSO Office	Final approved version — all 19 sections complete

Prepared by	Reviewed by	Approved by
EcoSphere Platform Team	Sustainability Manager	Chief Sustainability Officer
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

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